

Assessing Signatures of Climate Refugia and Future Range Shifts Using Bioinformatics and Genomic Tools

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Abstract

This capstone project integrates species distribution modeling (SDM), climate projection analysis, and genomic demographic inference to study climate refugia and future range shifts in selected Asian primates. The project focuses on *Trachypithecus phayrei*, *Trachypithecus pileatus*, *Trachypithecus geei*, and *Semnopithecus entellus*, using occurrence records, BIOCLIM environmental variables, past climate datasets, future CMIP6 scenarios, and SNP/VCF genomic data.

The SDM workflow uses MaxEnt to model habitat suitability from presence-only occurrence records. Occurrence data were cleaned, duplicate points were removed, and environmental predictors were selected through correlation analysis to reduce multicollinearity. A key correction in this final version is that the same selected BIO layers were used consistently for past, present, and future projections for each species. This makes the comparisons across time periods more reliable because habitat shifts are interpreted from changing climate conditions rather than from changing predictor sets.

The genomic component uses Multiple Sequentially Markovian Coalescent (MSMC) analysis to reconstruct historical effective population size from SNP/VCF data for *Trachypithecus geei*, *Trachypithecus phayrei*, and *Trachypithecus pileatus*. By combining ecological niche modeling with population history inference, this project identifies regions of stable suitability, potential climate refugia, and species-specific vulnerability under future climate change. The results support biodiversity conservation planning by connecting habitat suitability, historical demographic change, and projected climate-driven range shifts.

1. Introduction

Climate change is one of the major drivers of habitat alteration, range shifts, and biodiversity loss. Primate species with restricted ranges or specialized ecological requirements may be especially vulnerable when suitable habitat becomes fragmented or shifts away from current distributions. Understanding how habitat suitability has changed from past to present, and how it may change under future climate scenarios, is therefore important for identifying climate refugia and conservation priority zones.

This project examines Asian primates using a combined ecological and genomic approach. Species distribution modeling was used to estimate habitat suitability under past, present, and future climate conditions. In parallel, MSMC analysis was used to infer historical effective population size patterns from genomic SNP/VCF datasets. Together, these methods provide a stronger framework for understanding both where suitable habitats occur and how species populations may have responded to past environmental changes.

1.1 Objective

The objective of this project is to assess climate refugia, future range shifts, and historical population dynamics of selected Asian primate species by integrating SDM, MaxEnt ecological niche modeling, BIOCLIM climate variables, CMIP6 future projections, PaleoClim past layers, and MSMC genomic analysis.

1.2 Goal

The goal is to develop a reproducible analytical workflow that supports conservation planning by identifying stable habitat regions, potential future range shifts, and demographic patterns that may reflect past climate-driven changes.

2. Dataset and Methodology

2.1 Species Included

- *Trachypithecus phayrei*
- *Trachypithecus pileatus*
- *Trachypithecus geei*
- *Semnopithecus entellus*

2.2 Data Sources

- Species occurrence data: Global Biodiversity Information Facility (GBIF).
- Current climate data: WorldClim BIO1-BIO19 variables.
- Past climate data: PaleoClim layers representing Last Glacial Maximum, Pleistocene, and Pliocene periods.
- Future climate data: WorldClim CMIP6 projections under SSP126 and SSP585 scenarios.
- Genomic data: SNP/VCF files used for MSMC demographic analysis.

2.3 Data Preprocessing

Occurrence records were cleaned before modeling. Duplicate coordinates were removed, records outside the study region were filtered, and cleaned occurrence points were matched with environmental raster layers. Environmental predictor values were extracted from BIOCLIM rasters, and correlation analysis was performed to identify and remove highly correlated variables. This helped reduce multicollinearity and improved interpretability of the MaxEnt models.

2.4 BIOCLIM Layer Consistency

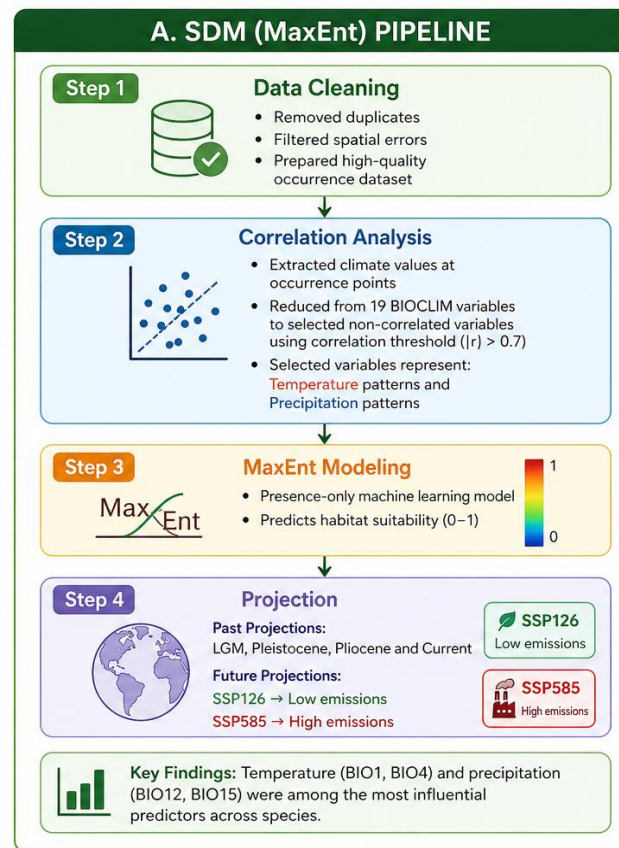
For the final modeling workflow, each species used the same selected BIO layers for past, present, and future projections. The selected variables were identified from the current climate correlation analysis and then applied consistently to the corresponding past and future raster sets. This correction is important because it keeps the predictor set stable across time periods, allowing the results to reflect climate-driven habitat changes rather than differences caused by using different input variables.

Species	Cleaned occurrence records	Same BIO layers used for past, present, and future projections
<i>Trachypithecus phayrei</i>	56	BIO1, BIO4, BIO8, BIO10, BIO12, BIO15
<i>Trachypithecus pileatus</i>	240	BIO1, BIO4, BIO12, BIO14, BIO15

Trachypithecus geei	44	BIO1, BIO4, BIO12
Semnopithecus entellus	258	BIO1, BIO4, BIO8, BIO9, BIO10, BIO12, BIO14, BIO15, BIO18, BIO19

2.5 SDM Workflow

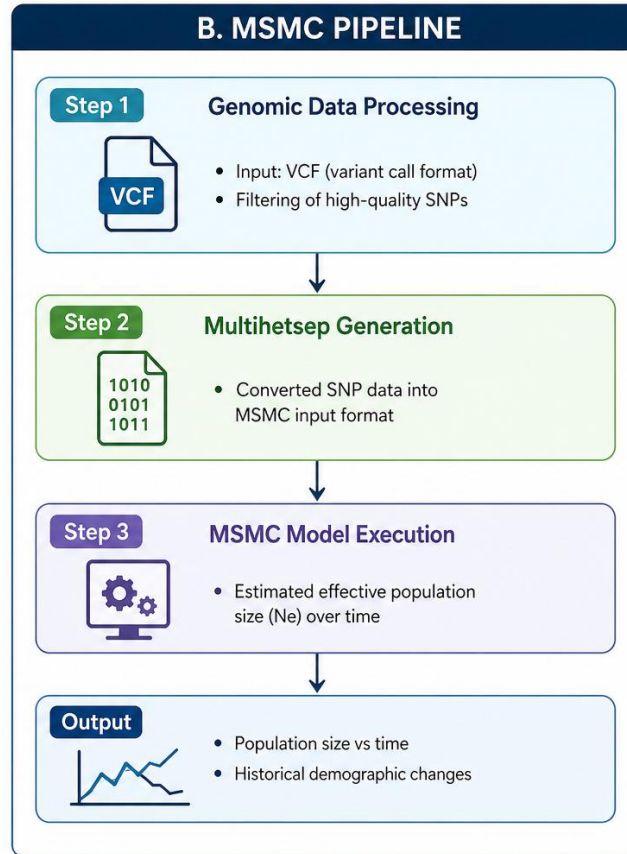
1. Clean GBIF occurrence records and remove duplicate coordinates.
2. Extract BIOCLIM values for occurrence locations.
3. Run correlation analysis to select non-highly correlated BIO variables.
4. Use the same selected BIO layers for past, current, and future projections for each species.
5. Crop and align environmental rasters to the study region.
6. Convert selected raster layers to ASCII format for MaxEnt.
7. Train MaxEnt models using current climate and presence-only occurrence points.
8. Project trained models onto past and future climate scenarios.
9. Generate habitat suitability maps, binary suitability outputs, and comparison figures.



2.6 MSMC Workflow

10. Prepare SNP/VCF genomic datasets for the target species.
11. Generate multihetsep input files using msmc-tools.
12. Run MSMC to infer historical effective population size through time.
13. Plot effective population size against years before present.

14. Interpret demographic trends alongside SDM-based habitat suitability patterns.



3. Model Development

The model development stage was divided into two linked analytical components: species distribution modeling using MaxEnt and demographic inference using MSMC. Unlike a standard supervised classification project, this capstone does not use logistic regression, random forest, XGBoost, or a neural network. The core model is MaxEnt for ecological niche modeling, supported by MSMC for genomic population-history inference.

3.1 MaxEnt Species Distribution Model Development

MaxEnt was selected because it is widely used for presence-only species occurrence data. The model estimates the probability of suitable environmental conditions by comparing the climate values at known occurrence points with the background environmental conditions across the study area. In this project, MaxEnt was trained using cleaned GBIF occurrence records and selected BIOCLIM predictors for each species.

For each species, the current climate layers were used for model training. After training, the same model structure and the same selected BIO layers were projected onto past and future climate datasets. This approach allowed habitat suitability to be compared across time periods while keeping the

environmental predictor set consistent. Using the same BIO layers across past, present, and future projections improved methodological consistency and reduced bias in range-shift interpretation.

MaxEnt input preparation

- Cleaned occurrence CSV files were prepared for each species.
- Selected BIOCLIM rasters were cropped to the study region and aligned to the same spatial resolution.
- Raster layers were converted to ASCII format because MaxEnt accepts ASCII environmental layers.
- Occurrence files and environmental layer folders were organized separately for current, past, and future modeling.

MaxEnt modeling settings

- Training data: cleaned GBIF occurrence points.
- Environmental predictors: selected BIOCLIM layers after correlation filtering.
- Projection periods: current climate, past climate periods, SSP126 future scenario, and SSP585 future scenario.
- Output type: habitat suitability maps and model performance outputs, including AUC values.
- Model interpretation: areas with higher suitability values were interpreted as climatically favorable habitats.

3.2 Climate Projection Model Development

Future projections were developed under two CMIP6 climate scenarios. SSP126 represents a low-emission pathway with stronger climate mitigation, while SSP585 represents a high-emission pathway with stronger warming and more severe climate change impacts. The trained MaxEnt models were projected onto both scenarios to evaluate how suitable habitat may shift under different future climate conditions.

Past projections were developed using PaleoClim datasets for older climatic periods such as the Last Glacial Maximum, Pleistocene, and Pliocene. These projections helped identify areas that may have remained suitable over long time periods. Regions that remained suitable across past and present projections were interpreted as possible climate refugia.

3.3 BIO Layer Selection by Species

BIO layer selection was performed separately for each species using correlation analysis. Highly correlated variables were removed to avoid redundancy, and the retained variables were used consistently across all time periods. The final selected variables represented a combination of annual climate patterns, seasonal temperature variation, and precipitation conditions.

3.3.1 BIOCLIM Variable Descriptions

The BIOCLIM variables (BIO1–BIO19) represent biologically meaningful climate variables derived from temperature and precipitation data. These variables are widely used in species distribution

modeling because they capture annual climate trends, seasonality, and environmental extremes that influence species habitat suitability.

Temperature Variables

- **BIO1 – Annual Mean Temperature**
Represents the average temperature across the entire year.
- **BIO2 – Mean Diurnal Range**
Represents the average difference between day and night temperatures.
- **BIO3 – Isothermality**
Calculated as the ratio of BIO2 to BIO7, representing how stable day-to-night temperature variation is relative to annual variation.
- **BIO4 – Temperature Seasonality**
Measures variation in temperature throughout the year using standard deviation.
- **BIO5 – Maximum Temperature of Warmest Month**
Represents the highest temperature recorded during the warmest month.
- **BIO6 – Minimum Temperature of Coldest Month**
Represents the lowest temperature recorded during the coldest month.
- **BIO7 – Temperature Annual Range**
Represents the difference between the warmest and coldest months (BIO5 – BIO6).
- **BIO8 – Mean Temperature of Wettest Quarter**
Represents the average temperature during the wettest three-month period.
- **BIO9 – Mean Temperature of Driest Quarter**
Represents the average temperature during the driest three-month period.
- **BIO10 – Mean Temperature of Warmest Quarter**
Represents the average temperature during the warmest three-month period.
- **BIO11 – Mean Temperature of Coldest Quarter**
Represents the average temperature during the coldest three-month period.

Precipitation Variables

- **BIO12 – Annual Precipitation**
Represents the total yearly rainfall.
- **BIO13 – Precipitation of Wettest Month**
Represents the highest monthly rainfall.
- **BIO14 – Precipitation of Driest Month**
Represents the lowest monthly rainfall.
- **BIO15 – Precipitation Seasonality**
Represents variation in rainfall across the year, measured using the coefficient of variation (CV%).
- **BIO16 – Precipitation of Wettest Quarter**
Represents total rainfall during the wettest three-month period.
- **BIO17 – Precipitation of Driest Quarter**
Represents total rainfall during the driest three-month period.
- **BIO18 – Precipitation of Warmest Quarter**
Represents rainfall during the warmest three-month period.
- **BIO19 – Precipitation of Coldest Quarter**
Represents rainfall during the coldest three-month period.

Species	Selected BIO layers	Ecological meaning of selected predictors
<i>Trachypithecus phayrei</i>	BIO1, BIO4, BIO8, BIO10, BIO12, BIO15	Annual mean temperature, temperature seasonality, temperature during wet/warm quarters, annual precipitation, and precipitation seasonality.
<i>Trachypithecus pileatus</i>	BIO1, BIO4, BIO12, BIO14, BIO15	Annual mean temperature, temperature seasonality, annual precipitation, driest-month precipitation, and precipitation seasonality.
<i>Trachypithecus geei</i>	BIO1, BIO4, BIO12	Annual mean temperature, temperature seasonality, and annual precipitation.
<i>Semnopithecus entellus</i>	BIO1, BIO4, BIO8, BIO9, BIO10, BIO12, BIO14, BIO15, BIO18, BIO19	Annual and seasonal temperature variables combined with annual, seasonal, warm-quarter, cold-quarter, and driest-month precipitation variables.

3.4 MSMC Model Development

MSMC analysis was used to reconstruct historical effective population size for *Trachypithecus geei*, *Trachypithecus phayrei*, and *Trachypithecus pileatus*. The SNP/VCF files were converted into multihetsep format using msmtc-tools, and MSMC was then run to infer population-size changes over time. This analysis complements the SDM results because demographic changes may reflect historical climate shifts, habitat contraction, expansion, or population fragmentation.

The MSMC outputs were visualized as effective population size through years before present. Species with larger and more stable effective population sizes were interpreted as having greater historical demographic stability, while species with smaller or declining population sizes were interpreted as potentially more vulnerable or historically restricted.

3.5 Integrated SDM-MSMC Framework

The final model framework connects ecological and genomic evidence. SDM outputs show where suitable habitats occur under past, current, and future climates. MSMC outputs show how population size changed through evolutionary time. When interpreted together, these methods provide stronger evidence for identifying climate refugia, evaluating vulnerability, and understanding whether historical demographic trends align with modeled habitat availability.

4. Results

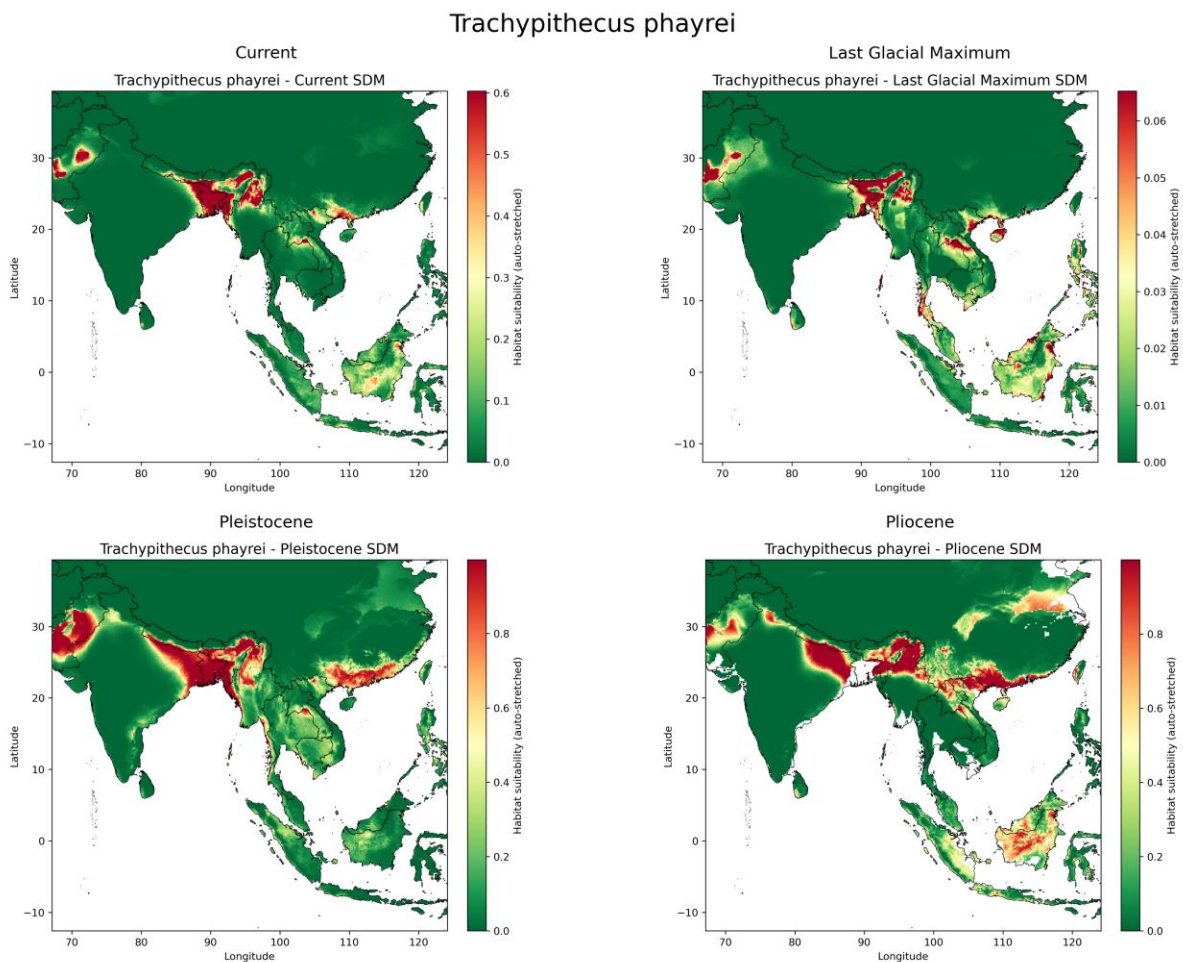
4.1 MaxEnt Model Performance

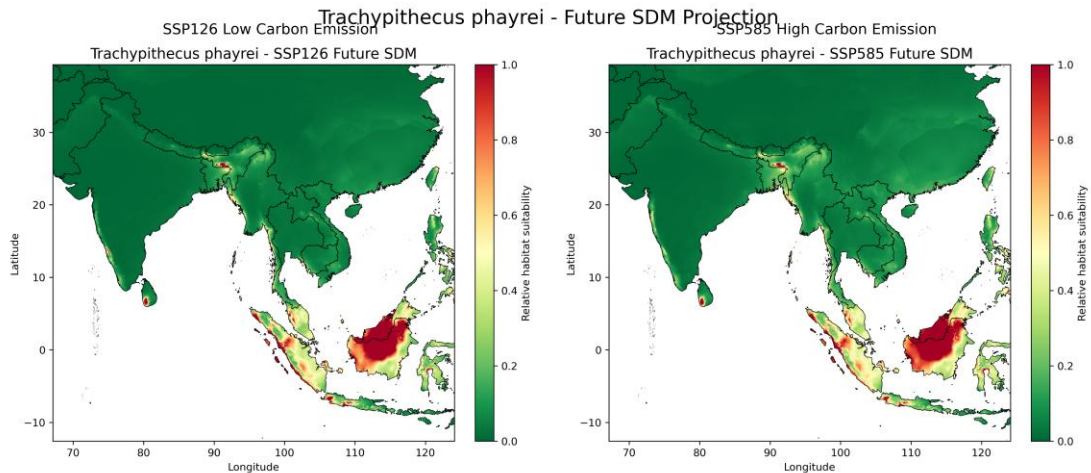
The MaxEnt models showed strong predictive performance across the species, with AUC values reported in the project repository around 0.87 to 0.91. These values indicate that the models were able to distinguish suitable from unsuitable environmental conditions with good accuracy. Because the same selected BIO layers were used across past, present, and future projections, the resulting maps provide a consistent comparison of habitat suitability over time.

4.2 Species Distribution Results

Trachypithecus phayrei

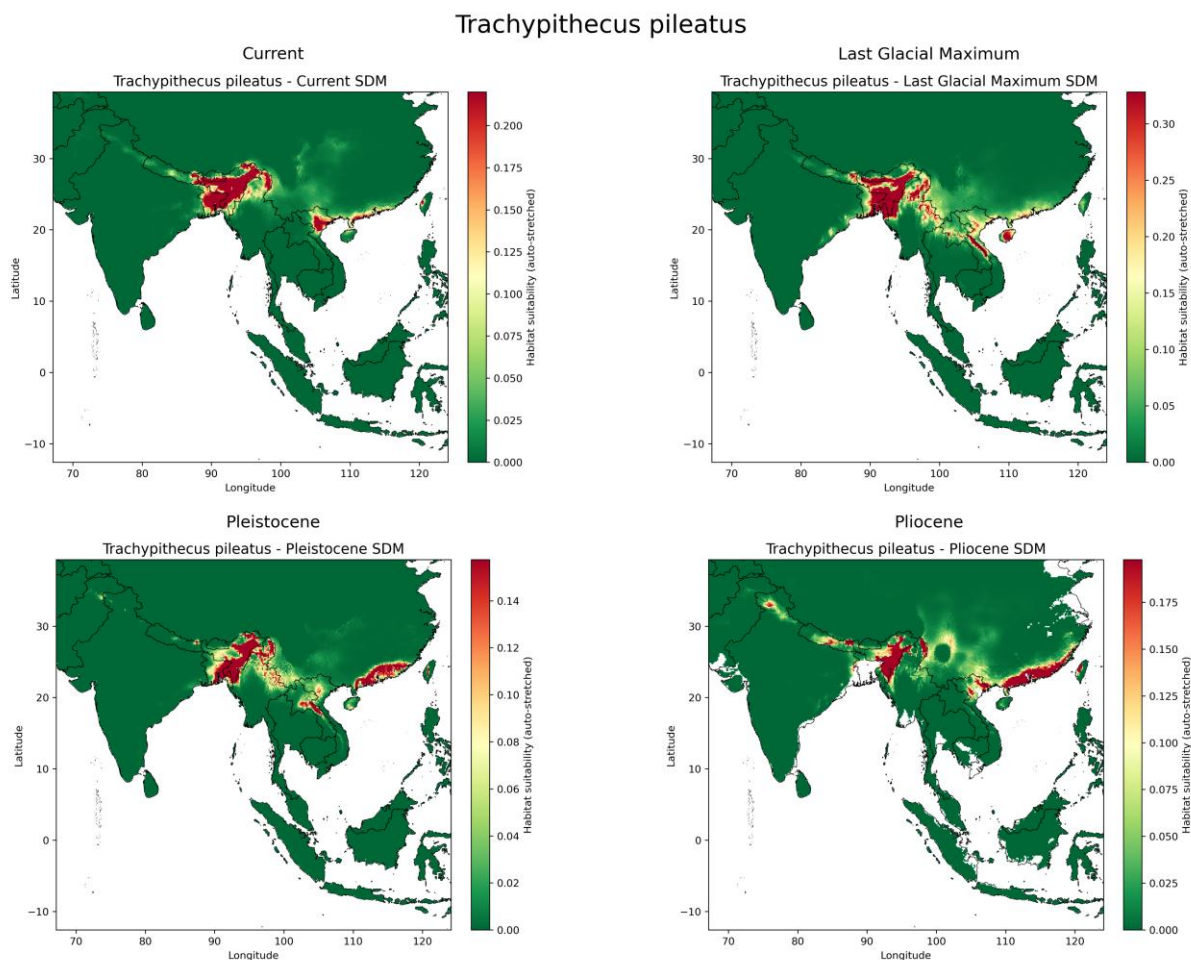
Trachypithecus phayrei showed strong habitat suitability in northeastern India, Bangladesh, Myanmar, and surrounding parts of mainland Southeast Asia. Future projections under SSP126 and SSP585 retained suitable areas in the same broad region, although SSP585 showed stronger fragmentation in some areas. Past projections suggested broader suitability during older climate periods, indicating that historical climatic conditions may have supported wider habitat connectivity.

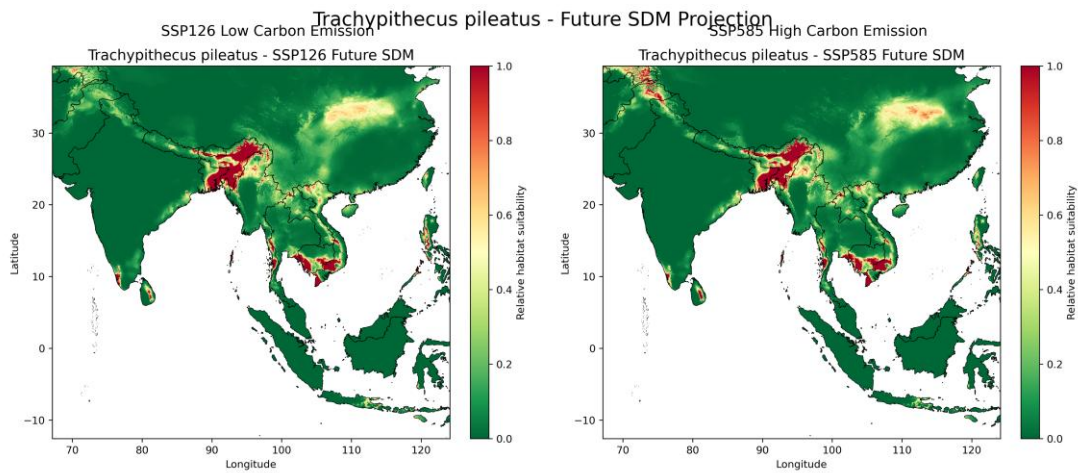




Trachypithecus pileatus

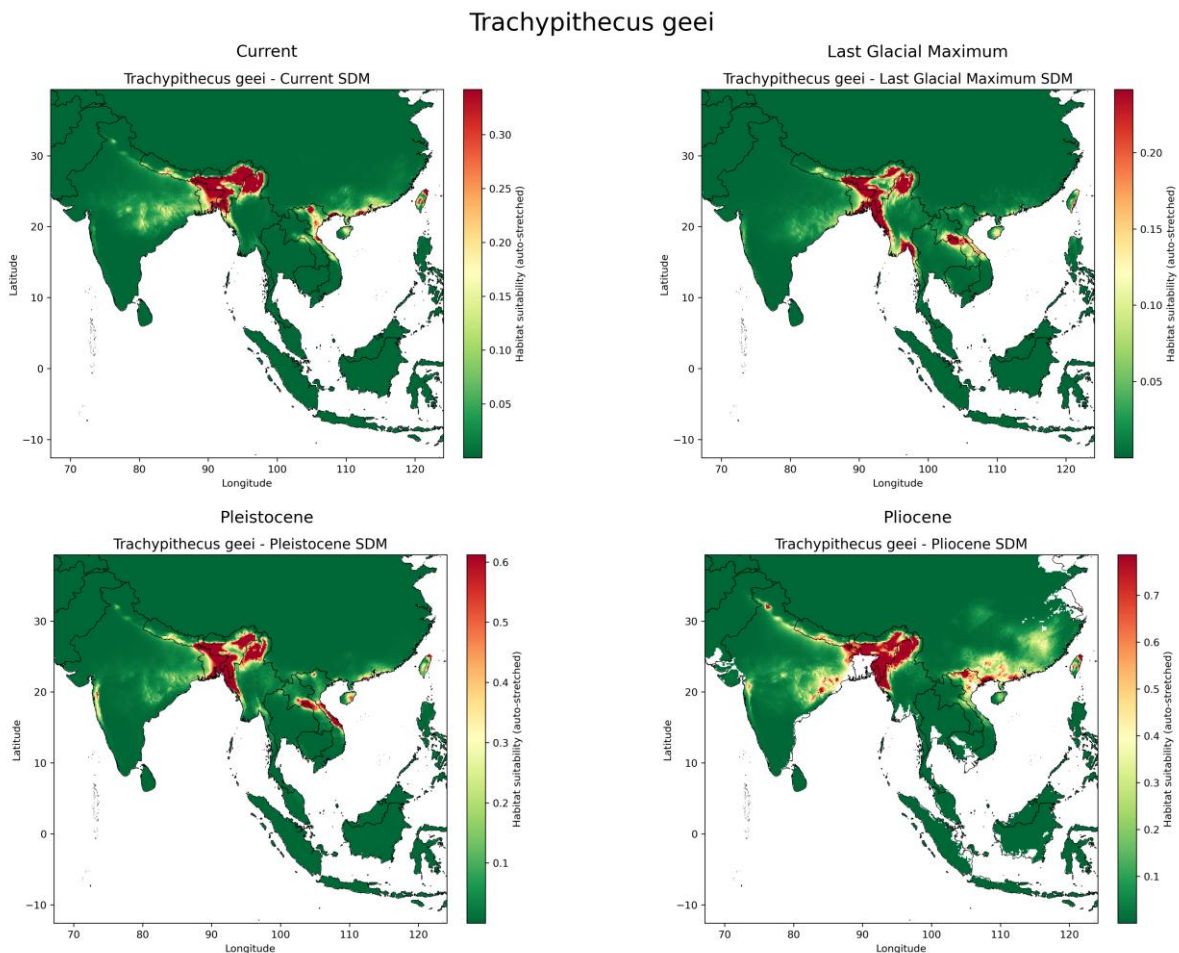
Trachypithecus pileatus showed high suitability across northeastern India, Bhutan, Bangladesh, and parts of Myanmar. Future projections under SSP126 and SSP585 were relatively similar, with suitable habitat remaining concentrated around northeastern India and the Himalayan foothill region. Past suitability patterns also remained broadly connected to this region, suggesting possible climatic stability in parts of its range.

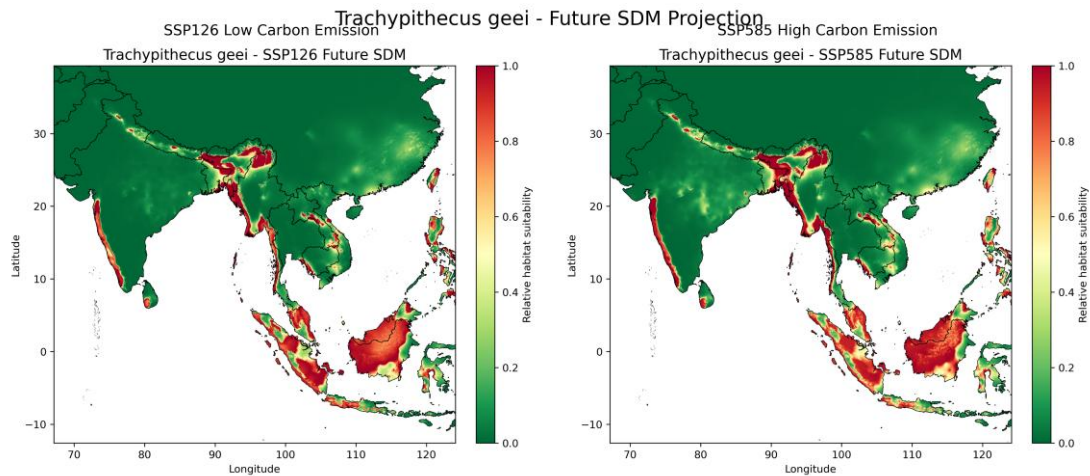




Trachypithecus geei

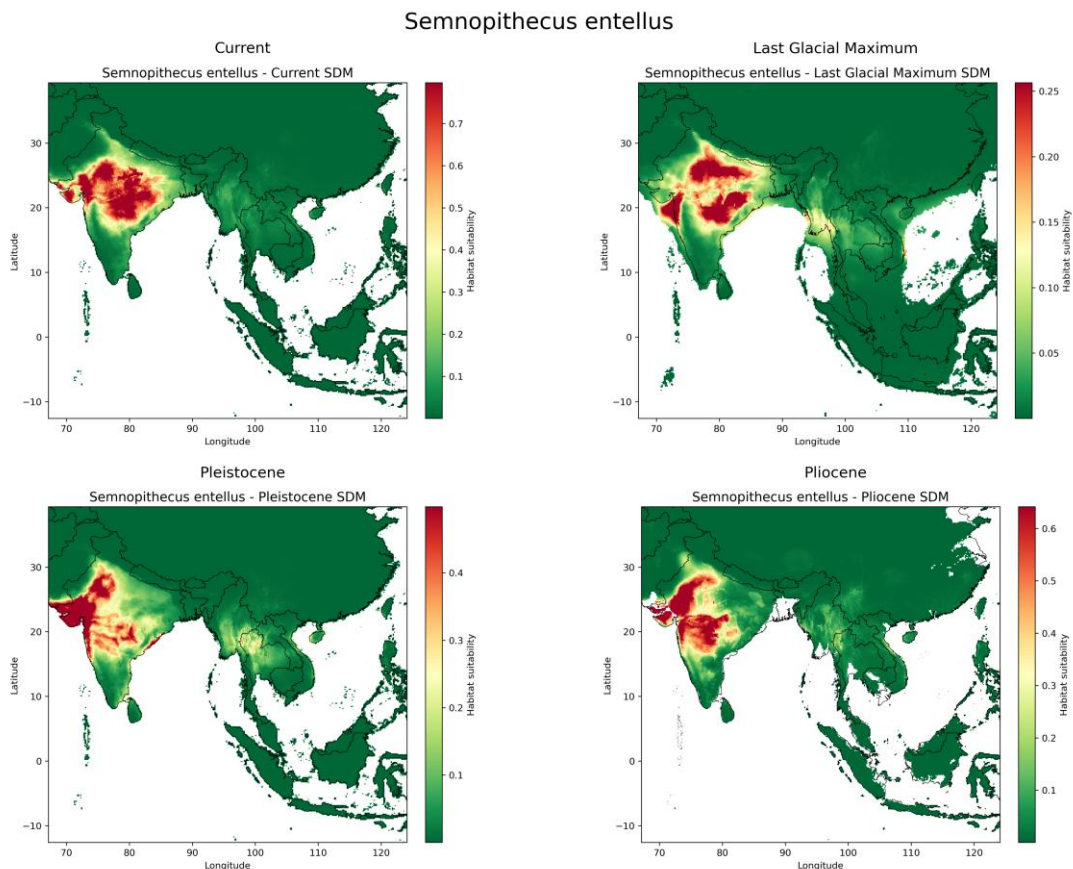
Trachypithecus geei showed strong suitability in northeastern India, Bhutan, and nearby transboundary regions. Compared with the other Trachypithecus species, its future suitable habitat appeared more restricted and vulnerable, especially under the high-emission SSP585 scenario. This suggests that T. geei may face greater risk from future climate-driven habitat narrowing and fragmentation.

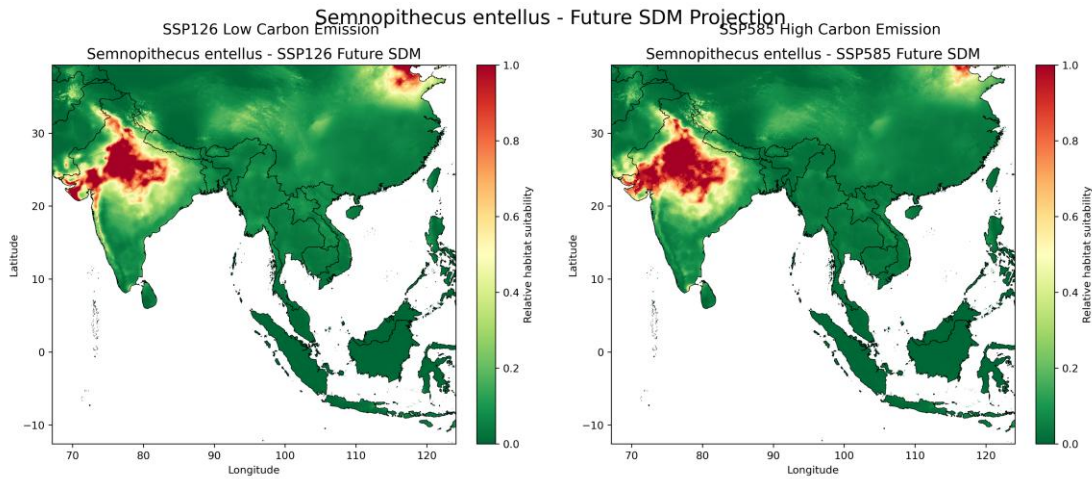




Semnopithecus entellus

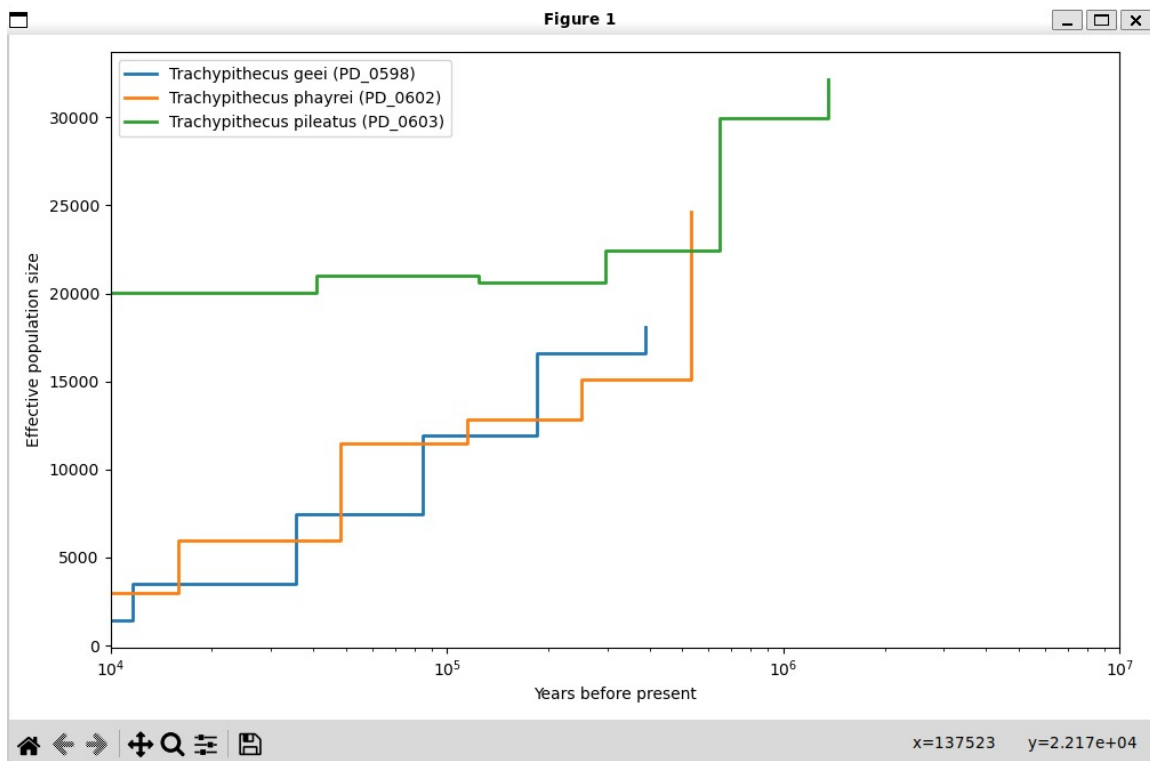
Semnopithecus entellus showed broader suitability across parts of India, especially western and central regions. Compared with the more range-restricted *Trachypithecus* species, *S. entellus* appeared to have a more stable and broader climatic niche. This pattern suggests that the species may be relatively more resilient under future climate scenarios, although localized habitat pressure and fragmentation may still affect conservation outcomes.





4.3 MSMC Results

The MSMC analysis showed different historical demographic patterns among the three *Trachypithecus* species. *Trachypithecus pileatus* maintained the largest effective population size across much of the inferred timeline, suggesting relatively higher historical demographic stability. *Trachypithecus phayrei* showed evidence of population-size change through time, with increases during older periods. *Trachypithecus geei* showed a smaller effective population size compared with the other species, supporting the interpretation that it may have experienced a more restricted or vulnerable demographic history.



4.4 Overall Result Summary

Analysis component	Main output	Interpretation
SDM / MaxEnt	Habitat suitability maps for past, present, and future scenarios	Identified suitable areas, potential refugia, and climate-driven range shifts.
BIO layer consistency	Same selected predictors used across time periods for each species	Improved fairness and reliability of past-present-future comparisons.
Future projections	SSP126 and SSP585 suitability maps	Compared low- and high-emission impacts on future habitat suitability.
MSMC	Historically effective population-size curves	Provided genomic evidence of species-specific demographic stability or vulnerability.

5. Discussion

The results show that combining SDM and MSMC provides a stronger understanding of climate refugia and future range shifts than using either method alone. MaxEnt identifies spatial patterns of habitat suitability, while MSMC provides demographic context about how populations changed through time. Together, these outputs help explain both ecological suitability and historical population response.

5.1 SDM Insights

The SDM results suggest that northeastern India, Bhutan, Myanmar, Bangladesh, and nearby transboundary regions are important for the *Trachypithecus* species. These areas repeatedly appeared as suitable in current and future projections, suggesting that they may include important conservation zones. However, future projections, especially under SSP585, showed possible habitat narrowing and fragmentation for some species. This was most notable for *Trachypithecus geei*, which appeared more vulnerable than *T. phayrei* and *T. pileatus*.

5.2 MSMC Insights

The MSMC results suggest that the three *Trachypithecus* species did not share identical demographic histories. *T. pileatus* showed larger historical effective population size, while *T. geei* showed smaller effective population size. This supports the idea that species with more restricted suitable habitats may also show more vulnerable demographic histories. These patterns can be used to prioritize species and regions for conservation attention.

5.3 Importance of Using the Same BIO Layers

Using the same selected BIO layers for past, present, and future modeling is one of the most important methodological corrections in this final report. If different BIO layers are used for different time periods, changes in suitability may partly reflect differences in model inputs rather than real climate-driven habitat shifts. By using the same selected predictors for each species across all periods, the final workflow provides a more valid comparison of climate suitability through time.

5.4 Conservation Implications

- Stable suitable regions across past and present projections may represent potential climate refugia.
- Areas that remain suitable under SSP126 and SSP585 may be important long-term conservation zones.
- Species with restricted future suitability, especially *Trachypithecus geei*, may require stronger conservation monitoring.
- Combining genomic and ecological evidence can improve conservation planning for climate-sensitive primates.

5.5 Limitations

- Occurrence records from GBIF may contain sampling bias and uneven geographic coverage.
- MaxEnt uses presence-only data, so absence records were not available for direct validation.
- Future climate projections depend on the selected GCM, scenario, and time period.
- MSMC results depend on VCF quality, filtering decisions, and available genomic samples.
- Land-use change, human disturbance, and dispersal barriers were not fully modeled.

5.6 Future Directions

- Include land-cover and human-impact variables to improve ecological realism.
- Compare multiple climate models instead of using only one future climate model.
- Add uncertainty maps for future suitability projections.
- Increase genomic sampling to strengthen MSMC demographic inference.
- Use ensemble SDM approaches to compare MaxEnt results with additional ecological niche models.

6. Conclusion

This project demonstrates how ecological niche modeling and genomic demographic inference can be combined to study climate refugia and future range shifts in Asian primates. MaxEnt SDM identified suitable habitats under past, present, and future climate conditions, while MSMC analysis provided historical population-size context. The final corrected workflow uses the same selected BIO layers for each species across past, present, and future projections, making the suitability comparisons more consistent and reliable.

Overall, *Trachypithecus geei* appears more vulnerable due to restricted future suitability and smaller inferred historical effective population size, while *Trachypithecus pileatus* and *Trachypithecus phayrei*

show broader or more stable suitability patterns. *Semnopithecus entellus* shows a comparatively broader climatic niche. These findings support the use of integrated SDM-MSMC workflows for conservation planning, especially in climate-sensitive biodiversity regions.

Code Availability

All project code, notebooks, outputs, and documentation are available in the GitHub repository: <https://github.com/AyeshaSiddiquag/Asian-Primate-SDM-MSMC-Capstone>

Approver Table

Approver Name	Title	Signature	Date
Dr. Kunal Arekar	Faculty Advisor / Mentor		05/07/2026
Srikanth Oruganti	Project Mentor / Reviewer		05/07/2026